

The CSC318 Survival Guide

I don't think writing notes for this course is very helpful since there's no final exam. You can always refer back to the slides. So I'll put this document here instead.

Can I skip classes? Are lectures recorded?

Attendance in the lectures are not graded, but you must be present for the studio sessions or risk losing marks. Therefore register for a time that doesn't conflict. For the lectures, you may attend sections that you aren't enrolled in, but your studios are stickier. However, for the purposes of finding teammates, you should attend the lecture you are enrolled in.

Do not expect these lectures to be recorded, although they can definitely be. I don't think Fanny and Khai record lectures, but Alex does.

Introduction and Project Examples

This is the closest to "the UI design course" offered by UofT's CS department. So, if you want to design user interfaces, this course might interest you. In this course, you work on a project that involves creating a *prototype* for a software that solves a real-world problem. That is, you identify a problem you see in the real world, and you *prototype* a piece of software that aims to solve it. That's the project.

Here are some thoughts:

- It's difficult to find teammates.
- I hate water fountains that give me IBS when I use them.
- I waste too much food.
- I want to know about more clubs.
- I want to know when the next bus is arriving.
- I don't want to smoke anymore.
- I want to sleep well.

Look out for project ideas you *do not* want to do. If you don't go to the gym, don't do a project related to the gym. If your teammates don't commute, don't make a transit app.

There are truly no prerequisites for this course. You do not need any programming experience, nor will it be any helpful. You just need to be good at writing.

Your final prototype will be on Figma. Think of making your prototype there as making a powerpoint that somehow has links for every button you have, but easier because Figma is actually intended for that.

Please do not take this course if you just want to learn how to make websites or UIs. Your time is better spent in CSC301 or CSC309. You're here to learn how to come up with good

ideas and execute solutions to them well. There are 10 lectures; 6 of them are for coming up with the idea for your application and 4 of them are for actually designing the UI for it. (That being said, 15% of your mark is on the former and 35% of your mark is on the latter)

What makes a project idea good?

PLEASE READ THROUGH THE ENTIRE DOCUMENT BEFORE MAKING A CONCLUSION ON WHAT PROJECT YOU WANT TO DO. Your idea should make A1, A2, and A3 easy for you.

Your project must solve a real-world problem. Sorry, making a Twitch extension is not a valid project.

It should be focused on one thing. What counts as a thing? Don't try to make an "everything" app; you will run out of time. Making an app to solve the issue of "stress" generically is considered **feature creep** and you will run out of time trying to solve every single problem. If you find yourself making two apps in one, drop one half and stick with the other. Here's an example of cutting a branch:

- A group makes a project that tracks the quality of computer keyboards and mice.
- This project is better off:
 - Focusing **completely** on one or the other but not both, OR
 - Abstracting away both into one singular categorical thing. Yes, the same type of abstraction you've learned in CSC207

Here are some of our scrapped ideas which would've been a nightmare to work with, so hopefully NO ONE comes up with these:

- An app that tries to get you engaged in politics by giving you facts and making you recite them just like how Duolingo does it (remember that being open has the caveat of a social taboo being marked as "we must accept it" moreover it's nearly impossible to be politically unbiased unless you're an election worker)
- An app that lets you manage your relationship with your roommates by assigning them tasks. (This will probably break friendships faster and I don't know how that will work).

Ideally a good project is where I can think of all the potential features with minimal ambiguities. For example, a platform that lets students report broken infrastructure is very clear and I know exactly how an app like that will work. However, an app that tries to condition a user to change their behavior might be less straightforward (or something your users might want to do, and rather see as propaganda). Unless you want to waste time in this course avoid un-straightforward ideas.

Your idea does not have to be unique. There are no bonus points to a unique idea. I think many of the ideas that were done have been done before. You have a lot of other courses to balance at the same time; do not overwhelm yourself.

Finding the right team

Your teammates must be in the same lecture section as you are. There are no exceptions.

Look for teammates with the same goals and work ethic. Moreover, I recommend getting teammates that are around your year of study. Many second-year students register for that course and you will not believe how difficult it is to work with them, so be very careful of letting them in unless they have completed an internship before. Also, be careful if your teammates are not in computer science.

Arguments can be made against teaming up with your friends. Work ethic and goals are not likely to align, and it's harder to hold them accountable. On the other hand, since you are more familiar with your friends, you *know* what to expect from them. Use that information when determining whether you want to pair up with your friends. Please note: just because they're in the same discord server with you does not make them the type of friend that is reliable - you need to get very familiar with them.

Goals and common interests are the most important. If your teammates have different goals than you, they might end up steering the project in a way that you might not desire the most, i.e. make suggestions that sound completely ridiculous. Don't be caught up in a scenario where a teammate thinks that making the background of an app highly saturated, adding distracting background music to a recorded presentation, or doing something when 500 red arrows point to your project to *not* do that (I had to wake up to multiple instances of the WCAG accessibility guidelines being violated and then having my teammate explain that it's a non-issue).

You can try finding 1-2 teammates before the course starts. That might help. Also, please don't rush, since the quality of your teammates matters. However, **if you do not find a team, you will be randomly assigned, but I think it's better to just drop the course by then.**

Please make sure at least ONE teammate who won't drop the course is a good speaker. Grammar matters here. Use a grammar checker (that isn't Grammarly).

Is this an easy 4.0?

Follow the instructions carefully and anything less than an 85% is completely your fault.

40% of your grade is individual. These are supposed to look like free marks; however, sometimes they are impossible to perfect just because of bad luck or impatience. If you're willing to put extra focus into the individual components by simply double-checking everything, I think scoring a perfect on this is possible. This makes this course very easy to pass.

Note: these components don't add up to 40%. The rest is free.

In-class activities (25%)

Activities you do during each lecture (except for 1). You will complete a worksheet and submit it. This is due at 9AM on the day **after** the lecture, hence attending the lecture is important. You will be given time during the lecture to complete it, but I don't think that's enough time.

Please read the rubric. Sometimes, it's hard to gauge what's enough. In this case, be extra descriptive and probably be prepared to sink an hour overthinking the process.

These can be graded harshly! Don't think that these are completely free!

Quizzes (4%)

A 10 minute quiz every week of lecture. Open book, but one attempt. Having the slides and abusing CTRL+F or other related tools is feasible; the questions are easy. Just remember to do them. This is usually due at 9AM after the last lecture of the week.

Studio Critiques (10%)

My only advice here: make them long and use complete sentences. If it's two sentences per point, that's not enough. Critiques that get full marks often consist of a document more than one page long.

The Group Project Phases

There are three phases to the group project (called assignments):

- The first one is on idea generation.
- The second one is on creating a paper prototype.
- The third one is on translating your paper prototype to a figma prototype.

There is a recorded presentation for you to make, although I don't think it's that difficult unless you're balancing way too many things at once.

Your assignments carry over (sort of). Rubric points on assignment 2 depend on promises you've made during assignment 1 (and so will assignment 3 for assignment 1 and 2) so keep this in mind. Knowing what the future assignments demand will help you.

Assignment 1

Your tasks are to:

- Identify the problem.
- Do a field study on the problem. Figure out what parts of the problem are making your stakeholders (people who benefit or would be impacted from your product) uneasy. You can meet this requirement by completing two of these for your primary stakeholders

(likely users of the app) **and** one of these for your secondary stakeholders (people who will be impacted by this app being used, but won't use it directly)

- Write a questionnaire. It's a survey that you would make on Google forms. Primary stakeholder requirement requires 15 responses, secondary stakeholder requirement requires 5 responses.
- Do an interview. Each interview lasts about 5-15 minutes, but this varies. Primary stakeholder requirement demands 2 interviews, secondary demands 1.
- Do an observation study. Requirements are the same as interviews. Looking at this, it seems infeasible and you should avoid this because it's nearly impossible to do.
- **Tip:** Having a project idea that ends up making course TAs secondary stakeholders is very helpful since your studio TA could be that interviewee and they'll be more than willing to be that interviewee
- Report some key findings.
- Report 5 design requirements. They are in the form "app should solve this problem, which is...", where you state the problem that is bugging off your stakeholders. **Do not state HOW to solve this problem**, this is for A2 onwards. **You should commit to solving these since your future assignments will depend on them.**

I would always recommend doing a questionnaire since getting the 15 responses is very easy if you spam them around Discord, especially if the topic is very relevant.

Please design your questionnaire and interview questions carefully. Since you want 5 design requirements, you want your questionnaires to end up giving you 5 separate problems. Don't end up with 4. Your questionnaire should help answer this question: "is this actually a problem?" or give you ideas for new problems.

Your results from your field studies **must** be frequently cited. Every single finding and design requirement must have evidence from your field study.

Use the open session as much as you can since that might be your only chance to get interviews.

All interviews must be recorded.

Assignment 2

A paper prototype is literally a prototype of your application on paper. "Honey I cannot click the book"

You will make a paper prototype of your application. It must be on paper. You must draw everything unless you are printing out a screenshot from Google maps. Whether you are allowed to digitally sketch parts of your prototype will be up to your TA.

You will also make storyboards, but that does not need to translate to what goes on the paper prototype. You can delegate one of your group members to do this instead.

You must then get **at least 4 other people NOT in your group** to test your paper prototype. These are called “think-alouds.”

You “test” your paper prototype by simulating it in front of them. Aim for your prototype to be at least 5-7 minutes long.

You must also get **4 other CSC318 students (or previous CSC318 students)** to perform a heuristic evaluation of your prototype as well as give feedback to it. These are called “heuristic evaluations.”

Then you must provide 3 findings based on the think-alouds you conducted **AND** 3 findings based on the heuristic evaluations, **for a total of 6**.

There are other parts, but you can refer to the assignment for that.

Your think-alouds will likely happen during the open session, but many groups choose to spend half of the time completing the prototype and the other half conducting think-alouds.

All think-alouds must be recorded.

Assignment 3

Make a final prototype on Figma.

It must be usable - buttons must work and redirect you to the final page, and it must look final. Anything that is off-center will cost you marks, so your UI better not look ugly.

Note: it is okay for a page to have 3 options, and regardless of what the user picks assume they have picked a fixed option.

You must specify how your prototype meets your design requirements. You may also come up with an excuse for not approaching that design requirement, or say that your prototype couldn't meet it. That is okay. See, that's why previous assignments matter.

You must also take the feedback you got from assignment 2 and **explicitly show (write down)** how you used that feedback (from your findings or related).

You will come up with a number of different ways to collect usability data. You can read the exact details on the assignment page. If you're reading ahead, don't worry too much about that for now.

That being said, here's an example of what I would do: Make a Google form that mimics the system usability scale questionnaire, **and** add your own questions as well. **You will want questions that ask: “did this feature solve the problem of ____”?** - This is VERY important.

Then conduct a usability study (your participant tests out your prototype and fills out the questions you have at the end). You must do this **8** times. Hence, have it done in parallel during the open session since doing it one at a time does not give you enough time.

You will then present 2 usability findings and 3 applicability findings, where:

- An example of a usability finding is that I should've used sliders instead of buttons. Or maybe that part of my app wasn't too intuitive.
- Applicability findings answer: "Does this solve this problem?" You fetch these findings by asking your participants directly, so please ask them so. Example: this feature did not solve anxiety because 90% of our participants said so (and why?)

There are other parts but I'll omit them here since they're straightforward.

All prototype walkthroughs must be recorded.

Assignment 4

This is the recorded presentation. This is due on the same day, or maybe earlier than assignment 3.

Follow the instructions. It is straightforward. Maybe delegate a group member to make and record the entire thing. Maybe don't do it if you put too much work in assignment 3.

The best presentations are played at the last lecture. One of them really made me laugh, but I think it's the sound of the incorrect buzzer when they tried a possible solution and it didn't work.

This one is very easy to get *ALMOST* full marks as long as you cover everything.

Studio Presentations

When an assignment is released, in the next studio you will do a presentation. Hence, it's possible that the assignment, or at least what you need to know will be released on the Thursday before the next presentation, which might be on next Tuesday. Hence, you don't have a lot of time. **For the first presentation you must have come up with the idea of your project.**

Do not overthink it. Read through the requirements, and get it done in some way. Stuttering is okay. Presentations are meant for feedback; you are judged for communicating your intentions, not how good they are.

The first presentation is probably the hardest to make; all the following ones are easier. Not all TAs enforce the 10-minute time limit, but don't take the risk.

Your slides do not need to look perfect. Just don't make them atrocious. These are marked leniently.

Overall tips

Read through the assignment requirements carefully. Always **verify** that you met all rubric points. Double check the gotchas, since the words “or” and “and” make a difference here. This document tries to cover all of them.

Please do not get AI to write for you. It is very easy to tell and no one wants to read content they can generate themselves.